EXAMINING TOTAL FACTOR PRODUCTIVITY IN INDIA'S UNINCORPORATED MANUFACTURING ENTERPRISES: STYLIZED FACTS AND DETERMINANTS

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ABSTRACT

What drives productivity in India's informal manufacturing sector? This study addresses this question by first estimating enterprise-level Total Factor Productivity (TFP) using a Value-Added Cobb-Douglas specification (VAKL) applied to unit-level data from the Annual Survey of Unincorporated Sector Enterprises (ASUSE), 2022–23. The dataset encompasses 115,406 manufacturing firms across 32 states and union territories. Subsequently, the study explores the determinants of TFP through multiple linear regression, focusing on four broad classifications of determinants: ownership characteristics, establishment characteristics, financial and technological status and the problems faced. The major contribution of the paper is the discussion on problems or obstacles faced by these establishments and its association with TFP, an area that has often been overlooked in existing literature. The study is essential since understanding the determinants of productivity in the unincorporated sector can help to improve the nation'ss inclusive growth trajectory.

Keywords: Total factor productivity, Informal Sector, Unincorporated enterprises, Firm size, Labour shortage

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

ASUSE	Annual Survey of Unincorporated Sector Enterprises
FSU	First Stage Unit
GVA	Gross Value Added
IMF	International Monetary Fund
IV	Instrumental Variable
NIC	National Industrial Classification
NSSO	National Sample Survey Office
OBC	Other Backward Classes
PIB	Press Information Bureau
PMMY	Pradhan Mantri Mudra Yojana
SC	Scheduled Castes
ST	Scheduled Tribes
TFP	Total Factor Productivity
UFS	Urban Frame Survey
VAKL	Value Added Specification Model

CHAPTER 1
INTRODUCTION

An unincorporated enterprise is a producer unit that is not incorporated as a legal entity separate from its owner (household, government or foreign resident); additionally, its owners are personally liable, without limit, for any debts or obligations incurred in the course of production (SNA, 1993). The informal sector is marked by less or no tax payment, less capital endowment, lower technology, less capital, lower wages and producing wage-goods compared to that of the formal sector (Gerxhani, 2004). In India, the unincorporated establishments refer to the non-agricultural enterprises, which are not registered under the Companies Act, 1956 or the Companies Act, 2013. The Companies Act, 1956 was introduced to regulate the formation, functioning and dissolution of companies in India. It provided a legal framework for company registration, shareholder rights and corporate governance. The Companies Act, 2013 replaced the earlier act and introduced significant reforms to achieve better governance, transparency, and accountability and thus, improve the ease of doing business in India.

The unincorporated non-agricultural sector plays a crucial role in the Indian economy contributing significantly to employment, Gross Domestic Product (GDP), and the overall socio-economic landscape. This sector not only sustains livelihoods for millions but also supports the incorporated sector by supplying goods and services, reinforcing its position in the domestic value chain (PIB, 2024). It absorbs a significant portion of the country's workforce, and is inclusive in providing employment opportunities to a diverse range of people, including those from rural areas and those with limited formal education. Between October 2023 and September 2024, this sector employed more than 12 crore workers with 7.34 crore establishments (PIB, 2025).

Given its scale and contribution, the unincorporated sector holds strategic importance in India's development landscape. To support this sector, the government rolled out various initiatives like Mudra Yojana and Startup India that aim to promote entrepreneurship and expand informal economic activities. The Annual Survey of Unincorporated Sector Enterprises (ASUSE) was conceptualized as a regular survey to provide data on different economic and operational characteristics of unincorporated non- agricultural establishments in manufacturing, trade and other services (excluding construction). Krugman (1994) famously stated, "productivity isn't everything, but, in the long run, it is almost everything." Total factor productivity (TFP) is a key economic indicator that reflects how efficiently inputs are being used to generate output. It represents the portion of output not explained by the amount of labour and capital employed. Statistically, TFP is measured as a residual—the part of a country's income that cannot be attributed to factor inputs such as labour and capital, which are easier to quantify (IMF).

This study aims to determine characteristics of the firms in India's unincorporated sector that influence their total factor productivity levels and to highlight the challenges they face. Two regression models have been estimated. In the first stage, firm-level TFP is calculated using the Value-Added Specification Model (VAKL), explained in Chapter 2. In the second stage, logarithm of TFP is used as the dependent variable and is regressed on 22 variables that capture different characteristics of the establishments. For simplicity, these variables are grouped into four categories: ownership characteristics, establishment characteristics, financial and technological progress, and problems faced.

The paper also provides a summary of the distribution of firms in unincorporated sectors across Indian states and union territories, and compares their mean TFP levels. Moreover, firms have also been categorised according to their National Industrial Classification (NIC) Code. The NIC is an essential statistical standard used to classify economic activities. For this study, NIC codes between 10-32, representing manufacturing firms under NIC-2008 (the latest classification), have been considered.

Thus, the paper explores the quality of the firms in the unincorporated sector and seeks to understand what drives their productivity. A key focus is to initiate a discussion around the often-overlooked problems faced by these establishments. Addressing these issues through targeted policies is essential for the better functioning of the sector. The relevance and contribution of the paper lie in the recognition that the informal sector serves as both, a source of jobs and a cushion for adverse economic shocks, as realised by policymakers in many countries (Maiti and Sen, 2010).

This paper is organized into five sections. The next section summarises the existing literature on total factor productivity, its connection with firm size and the informal sector. Section 3 describes the data and methodology, along with stylized facts. Section 4 presents the regression model and its results. Concluding remarks are provided in Section 5.

CHAPTER 2
THEORETICAL BACKGROUND

2.1 EXISTING LITERATURE

The literature on Total Factor Productivity (TFP) has traditionally focused on formal, incorporated firms, leaving a critical gap in understanding productivity dynamics within the unincorporated sector, despite its substantial contribution to employment and GDP in India.

A significant part of the literature has focused on the relation between firm size and productivity. De and Nagraj (2013) have used Levinsohn-Petrin approach to calculate TFP from the panel data they use; they find no evidence of firm size mobility lagging productivity. Posti and Maiti (2023) carried out similar research, for the informal sector of India and found a positive relationship between firm size and performance measure and evidence of labour exploitation. De Kok, Fris, and Brouwer (2006) find that although young firms start with below-average productivity and exhibit rapid catch-up growth, there is no systematic relationship between firm age and productivity growth beyond the first ten years of operation.

A related section of the literature focuses on gender disparity and eventual registration. Chaudhuri, Sasidharan, and Natarajan (2020) use a nationally representative dataset on India's micro, small, and medium enterprises to show that women-owned firms significantly underperform men's in size, growth, and efficiency, highlighting persistent gender bias in the credit market that, if addressed, could help narrow this performance gap. Kabuba-Mariara et al, 2023 conclude that for micro firms, registration is associated with between 49% and 65% increase in income.

An important paper in the limited literature available for the informal sector, is the paper by La Porta and Shleifer (2014), they dig into the informal sector, investigating sectoral size and reasons for its inefficiency. Banerjee and Duflo (2004) highlight firm-level heterogeneity, credit constraints, misallocation of resources, and market imperfections as key drivers impacting growth.

2.2 MODELING FRAMEWORK

My empirical framework starts with the Cobb Douglas Production function, Y= A f(L,K), which remains a widely adopted framework in empirical productivity analysis due to its clear economic interpretation. However, there are many assumptions to understand before it is applied, where A captures Hicks-neutral technological change, returns to scale are constant, and markets are assumed perfect (Orlando, 2023).

Using the Cobb Douglas Production function, we reach the calculation of TFP values. Here, TFP is calculated as follows (lowercase variables denote the logarithmic values of the corresponding uppercase variable):

$$va_i \;=\; \beta_0 \;+\; \beta_k\, k_i \;+\; \beta_\ell\, \ell_i \;+\; \varepsilon_i \tag{2.1}$$

where, firm-level value-added (va_i) is modeled as a function of inputs of capital (k_i) and labour (ℓ_i).

Firm's (logged) TFP is estimated as a sum of the constant and the residual, i.e.,

$$\mathrm{tfp}_i = \hat{\beta}_0 + \hat{\varepsilon}_i$$

The above model is referred to as value-added specification (VAKL) (Francis et al, 2020). However, there is a limitation attached with this model. The input endogeneity, where firms' input decisions correlate with unobserved efficiency—poses a challenge. Panel methods like Levinsohn-Petrin and Ackerberg-Caves-Frazer adjust for this in longitudinal data, but they are inapplicable to single-year cross-sections, forcing researchers to rely on the strong assumption of exogenous input choices or to seek valid instruments (Levinsohn & Petrin, 2002). An instrumental variable (IV) estimator achieves consistency by instrumenting the explanatory variables with regressors that are correlated with the inputs but uncorrelated with Eit. The IV approach can also alleviate measurement error problems, which tend to be most pronounced in capital. Potential instruments at the firm-level include input prices and lagged values of input use. Firm-level input prices are rarely observed (Levinsohn-Petrin, 2002). However, finding a suitable IV poses a challenge. Thus, an important assumption for this paper is, input choices are exogenous.

Even though unincorporated enterprises form the backbone of many developing-country economies, relatively few studies have examined their production efficiency or the unique challenges they face. This paper seeks to fill that gap by (1) estimating firms-level TFP for manufacturing establishments in India's unincorporated sector, and (2) highlighting the operational constraints: power supply, raw materials, finance, and labour, that have been largely overlooked in prior work.

CHAPTER 3
DATA, STYLIZED FACTS AND METHODOLOGY

3.1 DATA

The data used in this study are sourced from the Annual Survey of Unincorporated Sector Enterprises, 2022-23. The nationally representative survey is conducted yearly to collect information on the economic and operational characteristics of unincorporated non-agricultural establishments in the manufacturing, trade, and other services sector. The first comprehensive ASUSE was conducted by the National Sample Survey Office (NSSO) from April 2021 to March 2022. Previously, an all India survey on unincorporated non-agricultural enterprises was carried out in NSSO's 73rd round from July 2015 to June 2016. ASUSE was introduced to provide more frequent insights into this critical segment of the economy. For this study, the unit-level, ASCII-encoded raw data were extracted from ASUSE 2022–23.

The unit of enquiry of the ASUSE is an 'establishment'. An establishment is defined as an enterprise, or part of an enterprise, that is situated in a single location and in which either only a single productive activity is carried out or in which the principal productive activity accounts for major part of the value added. The total number of establishments in the sector were 6.50 crore in 2022-23. The survey has been conducted following a multi-stage stratified sampling scheme, where first stage units (FSUs) are census villages in rural and UFS (Urban Frame Survey) blocks in urban areas. In the ASUSE 2022-23, data were collected from a total of 4,58,938 establishments from 16,382 first stage units (FSUs). For the study, only the establishments with positive output were retained. The analysis focuses exclusively on manufacturing firms within the sector, classified under NIC codes 10 to 32 which make up 27.41% of the unincorporated sector. After cleaning and filtering, the final dataset contained 115,406 observations.

3.2 STYLIZED FACTS

Table 3.1 through Table 3.9 present simple bivariate comparisons of average TFP across key firm characteristics.

Male proprietorships dominate and report higher average TFP than female-led or partnership firms (Table 3.1). Most of the establishments (92%) employ five or fewer workers and exhibit the least productivity. Firms with 10-20 workers display the highest average TFP, though the relationship between size and productivity is non-linear (Table 3.2). Location matters, since, around 60% of the establishments operate from within the household premises and tend to be the least productive. Firms operating from fixed premises without a formal structure, though small in number, exhibit the highest productivity levels (Table 3.3). Demand shrinkage and non-recovery of financial dues emerge as the most frequently reported operational constraints, followed by erratic power supply, raw-material shortages, and labour limitations. Other challenges include erratic power, raw material shortage, and labour constraints (Table 3.4). These problems will be discussed in detail in Chapter 4. Most establishments are owned by individuals from the OBC category. ST owned firms report the lowest average productivity (Table 3.5), reflecting inequalities or limited access to resources. About 80% of the establishments have been in operation for more than three years and tend to be more productive than the newer establishments (Table 3.6), implying that experience matters. Only approximately 26% of the establishments are registered under any act/ authority and they turn out to be more productive (Table 3.7). There is huge variation in productivity by industry. The transport equipment manufacturing is one of the most productive industries while manufacturing of tobacco products lags behind (Table 3.8). States like Gujarat and Goa have the highest mean TFP while Uttar Pradesh and Madhya Pradesh report lower productivity (Table 3.9), thus showcasing regional disparities.

3.3 METHODOLOGY

The empirical strategy involves two stages. In the first stage, firm-level TFP is estimated using the VAKL approach (as discussed in Chapter 2) by regressing the logarithm of Gross Value Added (GVA) on the logarithms of capital and total workers employed. An important assumption here is that the input choices are exogenous. The constant and residual from this regression represent the logarithmic value of TFP. In the second stage, these estimated log TFP values serve as the dependent variable in a multiple linear regression that incorporates 22 explanatory variables. These indicators can be grouped under four broad categories: ownership characteristics, establishment characteristics, technological and financial factors, and the challenges faced by the establishment. The regression is estimated using the sample weights to ensure that the results are nationally representative.

S. No.	Ownership Type	Mean Relative TFP	Frequency %
1.	proprietary (male)	2.6228	54.31
2.	proprietary (female)	1.0000	44.26
3.	partnership with members of the same household	5.0031	0.65
4.	partnership between members not all from the same household	3.7481	0.72

Table 3.1: TFP by Ownership.

S. No.	Firm Size	Mean Relative TFP	Frequency %
1	≤ 5 workers	1.0000	92.80
2	5 to 10 workers	1.0646	3.41
3	10 to 20 workers	1.4680	2.75
4	> 20 workers	1.1427	1.04

Table 3.2: TFP by Firm Size

S. No.	Location	Mean Relative TFP	Frequency %
1	Within household	1.0000	59.01
2	Permanent Structure	2.1411	37.33
3	Temporary Structure/Kiosk/Stall	2.4521	1.30
4	Without Structure	3.1803	0.78
5	Mobile Market	2.2557	0.41
6	Street Vendors	2.8646	1.16

Table 3.3: TFP by Location of the Establishment

S. No.	Problem	% of Firms	% of Firms Reporting a Problem
1	Erratic power supply/power cuts	3.12	8.00
2	Shortage of raw material	2.28	5.86
3	Shrinkage/fall of demand	20.59	52.77
4	High cost/unavailability of credit	3.76	9.64
5	Non-recovery of financial dues	4.22	10.81
6	Labour unavailability	0.77	1.98
7	Skilled labour unavailability	0.75	1.93
8	Labour disputes	0.25	0.65
9	Others	3.26	8.34

Table 3.4: Major Problems Faced by Firms

S. No.	Social Group	Mean Relative TFP	Frequency %
1	ST	1.0000	5.10
2	SC	1.0699	13.01
3	OBC	1.1157	53.84
4	Others	1.5030	27.65
5	Not Known	1.0510	0.40

Table 3.5: TFP by Social Group of the Owner

S. No.	Years of Operation	Mean Relative TFP	Frequency %
1	Less than 1 year	1.0000	4.30
2	1 to 3 years	1.0840	14.73
3	More than 3 years	1.4667	80.97

Table 3.6: TFP by Number of Years of Operation

S. No.	Registration Status	Mean Relative TFP	Frequency %
1	Registered	2.0060	26.13
2	Not Registered	1.0000	73.87

Table 3.7: TFP by Registration Status

NIC Code	Industry	Mean Relative TFP	Frequency %
10	Food products	2.3362	16.36
11	Beverages	1.6653	0.78
12	Tobacco products	1.0000	4.26
13	Textiles	1.5878	8.13
14	Wearing apparel	1.4726	43.79
15	Leather products	2.4519	0.66
16	Wood and cork products	2.1877	3.83
17	Paper products	1.8081	0.67
18	Printing/media	2.4432	0.97
19	Coke/petroleum products	2.3664	0.06
20	Chemicals	2.0640	0.45
21	Pharmaceuticals	2.6787	0.04
22	Rubber/plastic products	2.8844	0.70
23	Non-metallic minerals	2.6257	3.60
24	Basic metals	3.3781	0.69
25	Fabricated metal products	2.8030	5.37
26	Electronics	2.5858	0.06
27	Electrical equipment	3.6606	0.30
28	Machinery	3.5319	0.37
29	Motor vehicles/trailers	3.4387	0.12
30	Other transport equipment	5.0570	0.05
31	Furniture	3.1484	4.41
32	Other manufacturing	5.2476	4.33

Table 3.8: TFP by Industry (NIC Code)

State Code	State	Mean Relative TFP	Frequency %
1	Jammu & Kashmir	1.2792	2.53
2	Himachal Pradesh	1.5522	1.32
3	Punjab	1.8998	2.94
4	Chandigarh	1.4497	0.13
5	Uttarakhand	1.8729	1.36
6	Haryana	1.6336	1.93
7	Delhi	2.1824	1.32
8	Rajasthan	2.1287	4.24
9	Uttar Pradesh	1.1812	10.09
10	Bihar	1.8447	4.60
11	Sikkim	1.6242	0.10
12	Arunachal Pradesh	2.0800	0.37
13	Nagaland	1.9442	0.32
14	Manipur	1.3449	1.19
15	Mizoram	3.0719	0.03
16	Tripura	1.9315	1.10
17	Meghalaya	1.4729	0.42
18	Assam	2.1467	1.41
19	West Bengal	1.3105	8.76
20	Jharkhand	1.2140	2.36
21	Odisha	1.1496	3.68
22	Chhattisgarh	1.1333	1.87
23	Madhya Pradesh	1.0000	6.79
24	Gujarat	3.2496	5.57
25	Dadra & Nagar Haveli & Daman & Diu	1.7410	0.22
27	Maharashtra	1.3035	7.31
28	Andhra Pradesh	1.5985	5.50
29	Karnataka	1.4642	4.86
30	Goa	2.4814	0.16
31	Lakshadweep	1.6780	0.10
32	Kerala	1.7688	4.61
33	Tamil Nadu	1.4975	8.49
34	Puducherry	1.7684	0.37
35	A & N Islands	1.2055	0.24
36	Telangana	1.3586	3.69
37	Ladakh	2.2099	0.10

Table 3.9: TFP by State

CHAPTER 4
REGRESSION AND RESULTS

4.1 THE MODEL

In the first stage model logarithmic value of GVA is the dependent variable. GVA is calculated by subtracting value of total inputs from value of total outputs. Following is the first stage model:

$$\log(\text{gva}) = \beta_0 + \beta_1 log(capital + \beta_2 log(labour)$$

Where,

log(capital): It is the logarithmic values of the value of capital. Capital here is the sum of total market value of hired assets and total market value of owned assets.

log(labour): It is the logarithmic value of the total number of workers hired by the establishment.

For the second stage, to estimate the effects of different characteristics of the establishment on its productivity, following regression is performed taking logarithmic value of TFP as the dependent variable and taking 22 indicators as independent variables, that includes control for state and industry.

$$
\begin{aligned}
\log(\text{tfp}) = \beta_0 &+ \beta_1 Nature + \beta_2 Years + \beta_3 NIC\_code + \beta_4 Computer + \beta_5 Internet \\
&+ \beta_6 RegistrationStatus + \beta_7 Ownership + \beta_8 LoanSource + \beta_9 FirmSize \\
&+ \beta_{10} State + \beta_{11} SocialGroup + \beta_{12} Location + \beta_{13} Education \\
&+ \boldsymbol{\theta}\textbf{Problems}
\end{aligned}
$$

Where,

Nature: Describes if the establishment operates perennially, seasonally, or casually. Perennial establishments are the base category.

Years: Represents the number of years the establishment has been operational. Three categories: less than one year, one to three years and more than three years. Less than one year is the base category.

NIC_code: Based on NIC 2008, takes values between 10–32 that indicate manufacturing firms.

Computer: Whether the establishment used computers in the last 365 days. Not using the computer has been taken as the base category.

Internet: Whether the establishment used the internet in the last 365 days. Not using the internet has been taken as the base category.

Registration Status: If a firm has been registered under any act/authority. Registered establishment has been taken as the base category.

Education: Represents the highest level of general education of the proprietor/major partner. Levels are: not literate, below primary, primary and above but below secondary, secondary and above but below higher secondary, higher secondary and above but below graduate and graduate and above. Not literate is the base category.

Ownership: Describes type of ownership. Proprietary (male), Proprietary (female), partnership with members of the same household, partnership between members not all from the same household and others. Male proprietary has been taken as the base category.

LoanSource: Three categories: loan taken from formal institutions, loan taken from informal institutions, or no loan taken. Loan taken from formal sources has been taken as the base category.

FirmSize: Based on the number of workers employed in the establishment. Categories: less than 5 workers, 5 to 10 workers, 10 to 20 workers, and more than 20 workers. Less than 5 workers is the base category.

SocialGroup: Social group of the proprietor/major partner. Categories: Scheduled Tribes (ST), Scheduled Caste (SC), Other Backward Classes (OBC), Others and Not known. ST is the base category.

Location: Describes where the establishment is located. Classifications: within household premises or outside household premises. There are five sub-classifications for outside household premises: fixed premises with permanent structure, fixed premises with temporary structure/kiosk/stall, fixed premises without any structure, mobile market, and without fixed premises (street vendors). Within household premises is the base category.

State: Control variable for states of India.

Problems: Vector includes the following dummy variables representing problems faced by the establishments:

- PowerProblem: Erratic power supply/power cuts. Base: No problem.
- RawMaterialProb: Shortage of raw material. Base: No problem.
- DemandProb: Shrinkage/fall in demand. Base: No problem.
- CreditProb: Non-availability/high cost of credit. Base: No problem.
- FinRecovery: Non-recovery of financial dues. Base: No problem.
- UnskilledLabourProb: Non-availability of labour as needed. Base: No problem.
- SkilledLabourProb: Non-availability of skilled labour. Base: No problem.
- LabourDisputeProb: Labour disputes and related issues. Base: No problem.
- OtherProb: Other unspecified problems. Base: No problem.

4.2 REGRESSION RESULTS

Table 4.1 below shows the regression results from model 1. Overall, 66.8% variation is captured by the model.

VARIABLES	(1) log_GVA	(2) log_GVA
log_capital	0.368***	0.368***
	(0.00263)	(0.00263)
log_labour	1.160***	1.160***
	(0.00459)	(0.00459)
Constant	4.197***	4.197***
	(0.0303)	(0.0303)
Observations	115,465	115,465
R-squared	0.668	0.668

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 4.1: Regression results for Model 1

Since the estimated model is a log-log model, the coefficients give the elasticity of GVA with respect to capital and labour. The capital elasticity is 0.368, implying that, ceteris paribus, a 1% increase in capital investment raises gross value added by 0.368%. Similarly, the labour elasticity is 1.160, indicating that a 1% increase in the number of workers hired, increases GVA by 1.160%, ceteris paribus. These results are significant at 1%.

Table 4.2 below shows the regression results obtained from the multiple linear regression. Overall, 28.9% variation is captured by the model.

VARIABLES	(1) log_tfp
Ownership (Ref: Male Proprietor)	
Female Proprietor	-0.529***
	(0.0125)

Table 4.2 continued.

	(1)
VARIABLES	log_tfp
Partnership- same HH	-1.117***
	(0.0328)
Partnership- Diff HH	-0.0563
	(0.0587)
Others	-0.160***
	(0.0543)
Operation nature (Ref: Perennial)	
Seasonal	-0.189***
	(0.0368)
Casual	-0.597***
	(0.133)
Number of years (Ref: less than one year)	
1 to 3 years	0.131***
	(0.0251)
More than 3 years	0.279***
	(0.0232)
Computer used (Ref: No)	
Yes	0.148***
	(0.0335)
Internet used (Ref: No)	
Yes	0.148***
	(0.0125)
Registration Status (Ref: Yes)	
No	-0.0233**
	(0.0119)
Education (Ref: Not literate)	
Below primary	0.0612***

Table 4.2 continued.

	(1)
VARIABLES	log_tfp
	(0.0200)
Primary and above, below secondary	0.0884***
	(0.0179)
Secondary and above, below higher secondary	0.106***
	(0.0184)
Higher secondary and above, below graduate	0.111***
	(0.0205)
Graduate and above	0.120***
	(0.0221)
Loan Source (Ref: Formal)	
Informal	0.0283
	(0.0278)
No access	-0.146***
	(0.0218)
Firm size (Ref: Less than 5 workers)	
> 5 workers but ≤ 10	-0.392***
	(0.0212)
> 10 workers but ≤ 20	-0.504***
	(0.0355)
More than 20 workers	-0.992***
	(0.0784)
Erratic power supply/ power cuts (Ref: No)	
Yes	-0.0182
	(0.0254)
Shortage of raw material (Ref: No)	
Yes	0.0683***
	(0.0242)

Table 4.2 continued.

	(1)
VARIABLES	log_tfp
Shrinkage/ fall of demand (Ref: No)	
Yes	-0.111***
	(0.0110)
Non- availability/ high cost of credit problem (Ref: No)	
Yes	-0.0521**
	(0.0257)
Non-recovery of financial dues (Ref: No)	
Yes	0.0366*
	(0.0189)
Non- availability of labour (Ref: No)	
Yes	-0.193***
	(0.0514)
Non- availability of skilled labour (Ref: No)	
Yes	-0.0614
	(0.0380)
Labour disputes (Ref: No)	
Yes	-0.180*
	(0.0997)
Other problems (Ref: No)	
Yes	-0.144***
	(0.0213)
Social group of the proprietor (Ref: ST)	
SC	0.0507*
	(0.0262)
OBC	0.0689***
	(0.0244)

Table 4.2 continued.

	(1)
VARIABLES	log_tfp
Others	0.0904***
	(0.0254)
Not known	0.0667
	(0.0566)
Location of establishment (Ref: Within household premises)	
Fixed premises, permanent structure	0.210***
	(0.0117)
Fixed premises, temporary structure	0.590***
	(0.0455)
Fixed premises, without any structure	0.709***
	(0.0714)
Mobile market	0.480***
	(0.0535)
Without fixed premises (street vendors)	0.671***
	(0.0351)
Constant	3.983***
	(0.0508)
State Controls	Yes
Industry Controls	Yes
Observations	115,406
R-squared	0.289

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 4.2: Regression results for Model 2

State and NIC codes are included as control variables. Results for other 20 variables have been explained below according to the four broad categories outlined earlier: ownership characteristics, establishment characteristics, financial and technological factors and operational challenges.

Ownership characteristics

Ownership characteristics include the type of ownership, the highest education level of the major partner/ proprietor, and the social group of the proprietor. Female-owned firms exhibit significantly lower productivity, ceteris paribus; it has 42% lower TFP compared to male proprietorships. Similarly, establishments with partnerships have lower productivity than male owned sole proprietorships. Thus, the results throw light on the existing gender productivity gap which demands policy attention. Previous research highlights that the intention and motivation of women entrepreneurs are different from male-led businesses (Chaudhari et al, 2018). Women-led businesses may be less interested in growth objectives but may be more motivated by the flexibility and the personal fulfillment it offers (Klapper and Parker 2011; Morris et al. 2006).

Social identity also plays a role. Compared to Scheduled Tribe (ST)-owned firms, those owned by Scheduled Castes (SC), Other Backward Classes (OBC), and 'Others' exhibit 5.1%, 6.9%, and 9.4% higher TFP, respectively. These disparities may reflect differential access to resources and opportunities.

Highest level of education obtained by the owner also plays a crucial role in determining the productivity of an establishment. Relative to the owners who are not literate, establishments led by owners with higher levels of general education are associated with significantly higher TFP. Ceteris Paribus, individuals with graduate-level education or higher exhibit 12.8% higher TFP, emphasising the importance of general education in enhancing efficiency.

Establishment characteristics

Establishment characteristics include the number of years the establishment has been operational, location, operation nature of the establishment, size, and the registration status.

Ceteris paribus, establishments operating for more than three years are 32.2% more productive than establishments that have been operational for less than one year. This implies that establishments are "learning by doing", the very act of operating can increase productivity since experience allows producers to identify opportunities for process improvements (Syverson, 2010). Taymaz (2002) also notes that new firms become aware of their actual productivity after observing their performance in the industry. Location of the establishment also plays a critical role. The establishments operating within household premises are the least productive. However, establishments operating from permanent but informally structured premises show a 103.2% higher TFP, followed by street vendors with 95.6% higher TFP, holding other factors constant.

Operational continuity is also essential to maintain better productivity. Ceteris paribus, seasonally operated firms are 17.3% less productive than perennial ones, and casually operating firms show a 45.1% decline in productivity.

Contrary to conventional expectations, the regression finds that increasing firm size, that is , the number of workers employed, is associated with a decline in productivity, potentially pointing to coordination problems or diminishing returns in informal setups.

Finally, registration status matters. Establishments that were registered under any Act/ Authority exhibit higher productivity compared to unregistered ones, possibly due to better access to institutional support or finance. Registration here means if the establishment is registered under any Act/Authority such as Shops and Establishment Act, Indian Trust Act (1882), Societies Registration Act, 1860, Co-operative Societies Act, 1912, Club or any other Act.

Financial and technological factors

Financial and technological factors include the loan source, computer use and internet access.

Access to loans impacts productivity. The establishments borrowing from informal sources do not have a significant variation from those that are borrowing from formal sources. However, establishments with no loan source, that is no access to loans, have 13.6% lower productivity than establishments borrowing from formal sources. Adopting the technology also provides gains. Ceteris paribus, establishments that use computers are significantly 16% more productive than those that do not use computers. Similarly, establishments with internet access also have 16% higher TFP. These results suggest that digital tools enhance information flow, market linkages, and planning capabilities, thereby boosting operational efficiency.

Problems

The survey captures a range of problems faced by the establishments, including erratic power supply, raw-material shortages, demand shrinkage, no or high cost credit, non-recovery of dues, unskilled and skilled-labour shortages, labour disputes, and other miscellaneous challenges.

According to the regression results, firms reporting a fall in demand suffer an 11.1% drop in productivity, emphasising the importance of a stable market. Competition from formalized goods (with warranties), the rise of e-commerce, and home-delivery services likely might be the reason behind this result. Establishments facing high-cost or unavailable credit have 5.2% lower productivity, ceteris paribus. This highlights the critical role of affordable, accessible financing in sustaining operations and facilitating investments in equipment or inputs. The most severe impact comes from a lack of unskilled labour to hire; a 19.3% decline in TFP, ceteris paribus. However, it is interesting to see that the lack of skilled labour does not have a statistically significant effect on TFP. Establishments facing labour disputes see an 18% lower productivity, ceteris paribus. This indicates that at small scale operations, conflicts can cause higher loss. Other problems that have not been captured by the survey also pose significant reduction in TFP values.

Surprisingly, keeping other things constant, raw material shortages and non-recovery of financial dues are associated with 6.8% and 3.7% higher TFP, respectively. It might be indicating that more productive firms face greater input demand and can better absorb payment delays. Erratic power supply does not have a significant impact on TFP.

CHAPTER 5
CONCLUSION

As noted in Chapter 1, the performance of the unincorporated sector is an essential aspect of overall economic development of India, yet its productivity drivers and operational hurdles remain poorly understood. This study leverages the newly released ASUSE 2022-23 dataset and follows a two-stage framework to expand the limited knowledge on the unincorporated sector. We find that older establishments tend to be more productive, and that both digital adoption and access to formal credit exert a highly positive influence on total factor productivity. In contrast, demand shrinkage, labour constraints and credit issues surface not only as the most prevalent challenges but also as the ones most harmful to firm-level performance.

To strengthen the functioning of this vital sector, targeted interventions are needed. First, digital training programs should be provided to equip the owners with the skills to to leverage computers and internet platforms for production and marketing, under the Skill India and Digital India programmes. Second, credit outreach must be broadened. This can be achieved by establishing a dedicated micro-finance institution or guarantee fund for unincorporated firms, and by expanding PMMY's low-interest windows, thereby lowering borrowing costs and improving access. Third, to address the most prevalent problem of demand shrinkage, the government can facilitate partnerships between organized retailers or e-commerce platforms and the informal producers, to ensure stability for the unincorporated sector. Fourth, to settle labour disputes faster, community level mediation centres can be set up. Similarly, some local cells can help connect the unemployed unskilled worker and the establishments looking for labour. Finally, for the infant establishments, that is, establishments that have been operating for less than one year, some subsidies or support schemes would be beneficial. This can enable them to survive the early phase and eventually mature since it is evident that older establishments are more productive. Further, field studies and surveys should be conducted, specifically to understand the troubles faced by them in detail. Thus, there remains much to do.

This study however, does face some limitations. The assumption that inputs are exogenous is quite a strong one. It would be more appropriate if a suitable IV is used for calculation of TFP. Moreover, conducting this analysis on a panel data would give more meaningful observations. Thus, by deepening the understanding of the micro level dynamics, future research can inform even more finely tuned policies to support India's unincorporated backbone and drive towards more inclusive economic growth.

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